

[Midterm Exam]

Fall 2022: CSE 309: Theory of Automata

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Sunday, October 16, 2022. Total Marks: 25, Duration: 120 minutes

– Solution –

Instructions

- Use of mobile devices is strictly prohibited for the entirety of the exam period.
- Please submit your devices in your bag at the front of the examination room.
- You may keep writing material with you on your desk.
- Please **do not** use a pencil or a red pen for answers that you want to be graded.
- Please write your answers in the **provided space**.
- Please **do not** tear/remove any page.
- Acquisition of answers through unfair means will automatically cancel your exam.
- Keep track of the time.
- Please write your **name** and **student ID** clearly on all pages you submit.
- This exam contains **5 questions** for a total of **25 points on 6 sides** including this.

Question 1

[5 points]

Prove/disprove the following Proposition:

Proposition. *Any subset of a regular language is regular.*

Solution

The given Proposition is *false*

Counterexample. Let $A = \{0^*1^*\}$. Clearly A is regular (we can create a DFA/NFA for A). Now consider the language $B = \{0^n1^n : n \geq 0\}$. The language B is a subset of the language A . However, we know B is not regular. Therefore the given proposition is *false*.

Question 2

[5 points]

Let $G = (V, \Sigma, R, S)$ be the grammar where $V = \{S, T, U\}$, $\Sigma = \{0, \#\}$, and R is the following set of rules:

$$S \rightarrow TT \mid U$$

$$T \rightarrow 0T \mid T0 \mid \#$$

$$U \rightarrow 0U00 \mid \#$$

- (a) (2 points) Give two strings generated by the above grammar.
(b) (3 points) Describe $L(G)$.

Solution

(a) $\#$, $\#\#$, $0\#0\#0$, $00\#0000$, etc.

(b) There are two different kind of strings in this language, either there is only one $\#$ sign such that all strings are of the form $0^n\#0^{2n}$ or there are exactly two $\#$ signs separating arbitrary number of 0's. In other words:

$$L(G) = \{0^n\#0^{2n} : n \geq 0\} \cup \{0^*\#0^*\#0^*\}.$$

Question 3

[5 points]

Show that following language A over the alphabet $\Sigma = \{0, 1\}$ is not regular using the *pumping lemma for the regular languages*:

$$A = \{0^k w 0^k : k \geq 1, w \in \Sigma^*\}.$$

Solution

Proof. Assume the language A is regular.

Let p be the pumping length for A . Consider the string $s = 0^p w 0^p$. Clearly $|s| \geq p$. Let us divide s into three parts as following: $s = xyz$, such that $x = 0^{p-1}$, $y = 0$, and $z = w 0^p$. Since $|y| > 0$ and $|xy| \leq p$, the two conditions of pumping lemma are satisfied. Let us consider the first condition i.e., for each $i \geq 0$, $xy^i z \in A$. Which is clearly not true because *pumping up* will increase the number of 0's on the left part and *pumping down* will decrease the number of 0's in any case the resulting string will not be in the language. Therefore, the *pumping lemma* does not hold for the language A and our assumption that A is regular is wrong.

Question 4

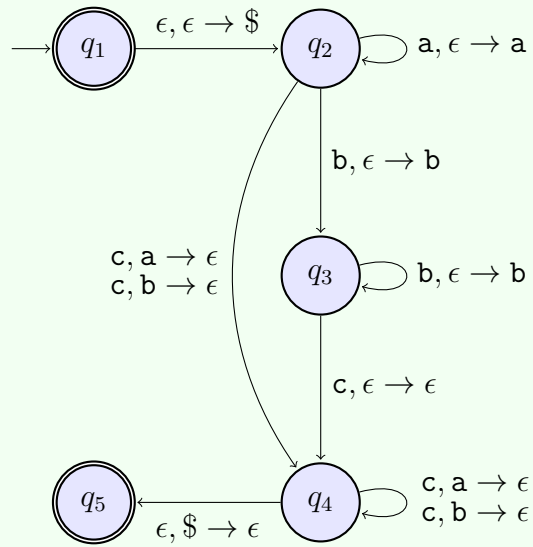
[5 points]

Design a *push down automaton* for the language B over the alphabet $\Sigma = \{a, b, c\}$:

$$B = \{a^n b^m c^k : k = n + m\}.$$

Solution

Assuming $n, m, k \geq 0$. Following PDA will accept the language B .



Question 5

[5 points]

Design a Turing machine for the following language C over the alphabet $\Sigma = \{0, 1, 2\}$:

$$C = \{0^i 1^j 2^k : i, j, k \geq 0, i = j = k\}.$$

[Note: You may provide your solution as a description or represent it graphically.]

Solution

Following Turing machine M_c decides the language C .

$M_c =$ “On input string w :

1. Scan the input from left to right to determine whether it is of the form $0^*1^*2^*$ and reject if it isn't.
2. Return the head to left-hand end of the tape.
3. Cross off a 0 and scan to the right until a 1 occurs and then scan to the right until a 2 occurs. Shuttle between 0's, 1's and 2's, crossing off one of each. If all 0's, 1's and 2's are gone, *accept*. If some 0's, 1's or 2's are left, *reject*.”